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Multivariable Current Controller for Enhancing Dynamic Response and Grid Synchronization Stability of IBRs

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ABSTRACT

This paper develops a multivariable current control strategy for inverter-based resources (IBRs) based on optimal control theory to enhance their dynamic performance and grid synchronization stability. The structure of the implemented multiple-input, multiple-output (MIMO) controller closely resembles that of the commonly used conventional single-input, single-output (SISO) PI controllers for IBRs. As a result, it requires only minor adjustments to conventional vector current control schemes, thereby facilitating its straightforward adoption. Time-domain simulations and analytical analysis demonstrate the superior performance of the developed method under various conditions and use case scenarios, such as weak power systems and uncertain parameters.

1 | Introduction

Power grids are adopting a higher share of inverter-based resources (IBR), such as wind and solar power generation and battery energy storage systems. As a result, the dynamics of IBRs are becoming influential on the overall power grid dynamic responses, making it crucial to enhance the IBR controller's dynamics [1–3].

One of the most common control strategies for the voltage source converter (VSC) is vector control in the synchronous reference frame (SRF) [4]. The model of the system in the dq frame constitutes a MIMO system [5]. The d and q axes are decoupled in conventional controllers using ideal proportional decoupling terms. Then, SISO PI controllers are used to control each axis. This control approach is effective and easy to implement, making it a popular controller for VSCs. However, it has limitations regarding dynamic performance and grid synchronization stability.

Due to various neglected dynamics in designing the decoupling terms in conventional controllers, such as delays in different components of the inverter and the phase-locked loop (PLL) dynamics, the two axes are not fully decoupled during transients. During disturbances, especially in weak grid conditions, the response of one axis is affected by the other. This deteriorates the dynamic response of the VSC when facing disturbances in the power grid or changes in the controller's set points [6].

Several attempts have been made to address the above challenges. Complex transfer functions were initially used for designing induction motors' current controllers [7, 8]. The VSC's transfer function symmetry is leveraged in the complex transfer function to convert the original MIMO transfer functions into SISO transfer functions. In this way, a conventional SISO controller can be used for the VSC, providing better responses than directly using conventional ideal decoupling terms [9, 10]. However, utilizing a complex transfer function restricts the degree of freedom in controller design, as the controllers for different axes of the

Abbreviations: IBR, Inverter-based resource; LQR, Linear quadratic regulator; MIMO, Multi-input multi-output; PLL, Phase-locked loop; SISO, Single-input single-output; SCR, Short circuit ratio; VSC, Voltage source converter.

original MIMO system cannot be designed independently. Also, direct cancellation of zeros of controllers and poles of the VSCs is prone to numerical inaccuracies and model uncertainties.

In [11–13], a state feedback MIMO controller is developed. In [12], the performance is enhanced by incorporating the PLL dynamics in the controller design process. Accessing the full state of the system may require additional state observers, which complicates the control design. In [13], PLL dynamics are also added to the state space model of the system used for designing the controller. Such explicit modeling of the PLL dynamic in the state space requires information about the power grid Thevenin equivalent parameters, which is a challenging task to obtain. The Thevenin equivalent may also change over time due to changes in the power grid's operating point and/or topology.

Nonlinear controllers such as feedback linearization and sliding mode controllers [14] and Lyapunov-based controller design and stability analysis [15, 16] have also been proposed to enhance the responses of VSC current controllers. However, such methods do not preserve the structure of the conventional VSC, which increases the complexity of the control design, tuning, and studying system impacts.

Different control parameter tuning strategies have also been proposed, developing trade-offs among controller objectives. For instance, [17] recommends reducing the bandwidth of controller parameters to enhance the grid synchronization stability. Although this method reduces the interactions between the control loops, it may lead to a sluggish controller that cannot track the fast-changing condition of the system [18, 19].

This paper implements a MIMO current control approach based on optimal PI control theory to enhance the response of VSCs with the following features:

- The structure of the controller resembles that of the commonly used conventional PI-based VSC controllers. It does not add any complexity to the controller structure as it preserves the structure of conventional controllers while offering superior performance.
- The controller parameter tuning is systematic and utilizes optimal control design using linear quadratic regulator (LQR), which is a well-established control synthesis method. As such, it has a classical robustness property (i.e., gain and phase margin) associated with the LQR design methodology.
- The controller enhances the dynamic response of VSCs by providing a faster response, less overshoot, and faster settling time.
- The controller improves the grid synchronization stability of VSCs. This is demonstrated through time-domain simulations and analytical analysis, where it does not require rigorous tuning as the grid strength changes [17].

The rest of the paper is organized as follows: Section 2 briefly explains the structure of conventional SISO current controllers. Section 3 explains the developed current controller for VSCs. Section 4 presents time-domain results and analytical analysis that demonstrate enhanced dynamic performance and grid synchronization stability of the controller. Finally, Section 5 concludes the paper.

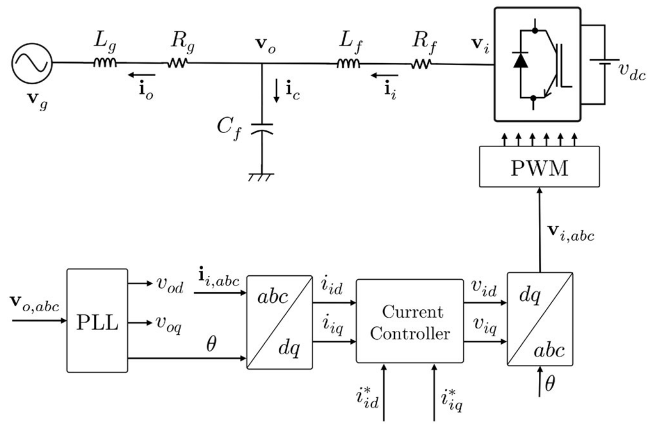

FIGURE 1 | Schematics of a typical controller of VSC.

2 | Conventional SISO Current Controller

Figure 1 shows a typical grid connected IBR. By applying Kirchhoff's Voltage Law (KVL) to the filter, the following can be derived:

$$\frac{di_{id}}{dt} = \frac{1}{L_f}(v_{id} - R_f i_{id} + \omega L_f i_{iq} - v_{od}) \tag{1}$$

$$\frac{di_{iq}}{dt} = \frac{1}{L_f}(v_{iq} - R_f i_{iq} - \omega L_f i_{id} - v_{oq}) \tag{2}$$

In (1) and (2), v_{id} and v_{id} are input variables and i_{id} and i_{iq} are output variables. The VSC controller determines the values of v_{id} and v_{iq} such that i_{id} and i_{iq} become equal to the desired values determined by the set points i^*_{id} and i^*_{iq}.

According to (1), the state space in d-axis is coupled with that of q-axis due to the $\omega L_f i_{iq}$ term. Similarly, according to (2), the state space in q-axis is coupled with that of d-axis due to the $-\omega L_f i_{id}$ term. In conventional vector controllers of VSC, these two terms are measured and provided to the controllers of each axis to cancel out their impacts. Once two axes are decoupled, conventional SISO PI can be used to control each axis. Note that v_{od} and v_{oq} are also measured and are provided as feedback to the controller to cancel out their impacts. Figure 2 shows the overall schematic of the conventional current controller.

Using the above-mentioned ideal decoupling terms (i.e., $-\omega_s L_f i_{iq}$ and $\omega_s L_f i_{iq}$, where ω_s is the synchronous frequency) has an acceptable steady-state response, but it deteriorates the transient response of the controller. This is because, due to delays in the measurement filters, PWM, and the dynamic response of the PLL or any other grid synchronization mechanisms, the ideal cancellation of the coupling terms is impossible.

3 | MIMO Current Controller for VSC

This section explains the developed current controller for VSCs, which is based on optimal control theory [20, 21]. The state space

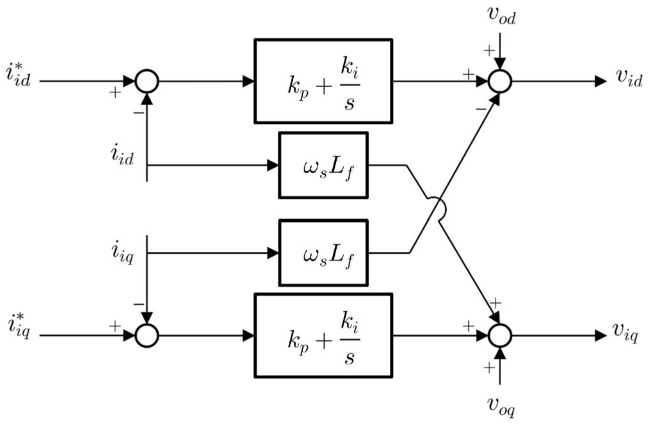

FIGURE 2 | Conventional SISO-PI based current controller of VSCs.

model in (1) and (2) can be represented as follows,

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\,\mathbf{x} \end{aligned} \tag{3}$$

where $\mathbf{A} = \begin{bmatrix} -\frac{R_f}{L_f} & \omega \\ -\omega & -\frac{R_f}{L_f} \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} \frac{1}{L_f} & 0 \\ 0 & \frac{1}{L_f} \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. In this representation, $\mathbf{x} = \mathbf{y} = [\, i_{id},\, i_{iq}\,]^\top$ and $\mathbf{u} = [\, v_{id},\, v_{iq}\,]^\top$. The objective of the controller is to regulate the output current to the desired setpoint $\mathbf{x}^* = \mathbf{y}^* = [\, i^*_{id},\, i^*_{iq}\,]^\top$. Since the objective of an LQR controller is to regulate the system states to zero, a shifted dynamic model of the system must be derived when the controller is intended to regulate the states to nonzero values. This approach contrasts with integral-action-augmented state feedback controllers designed via pole placement [22], in which the controller design is carried out directly using the original state space model, without any model shifting. Assume that the corresponding steady-state values of the state and input variables at the desired setpoint $\mathbf{y}^*$ are denoted by $\mathbf{x}^*$. In general, unlike $\mathbf{y}^*$, which is known prior to controller design, the values of $\mathbf{x}^*$ and $\mathbf{u}^*$ are initially unknown and can be determined after the controller has been designed. However, since the output matrix $\mathbf{C}$ in (3) is the identity matrix, it follows directly that $\mathbf{x}^* = \mathbf{y}^*$.

At steady state, the constant input $\mathbf{u}^*$ must satisfy (4). To keep the system states at a desired setpoint $\mathbf{x}^*$, one needs a constant input $\mathbf{u}^*$ that satisfies (4).

$$\mathbf{0} = \mathbf{A}\,\mathbf{x}^* + \mathbf{B}\,\mathbf{u}^* \tag{4}$$

Since $\mathbf{B}$ is invertible, $\mathbf{u}^*$ is calculated as (5).

$$\mathbf{u}^* = -\,\mathbf{B}^{-1}\mathbf{A}\,\mathbf{x}^* \tag{5}$$

As discussed above, this paper explicitly utilizes the facts that the current dynamics of VSCs are represented by the state-space model in (3), where $\mathbf{B}$ is invertible, and $\mathbf{C}$ is an identity matrix. This makes the resulting control structure resemble the conventional PI-based controllers commonly used in VSCs. According to [23], the error terms in the input, states, and the output of the system are defined as in (6) and (7), respectively.

$$\mathbf{e}_x = \mathbf{e}_y = \mathbf{x} - \mathbf{x}^* = \begin{bmatrix} i_{id} - i^*_{id} \\ i_{iq} - i^*_{iq} \end{bmatrix} \tag{6}$$

$$\mathbf{e}_u = \mathbf{u} - \mathbf{u}^* = \begin{bmatrix} v_{id} - v^*_{id} \\ v_{iq} - v^*_{iq} \end{bmatrix} \tag{7}$$

The integral terms in the PI controller are introduced as new states called $\mathbf{z}(t)$ as in (8).

$$\mathbf{z}(t) = \int_0^t [\mathbf{x}(\tau) - \mathbf{x}^*]\, d\tau \tag{8}$$

A new state space with the addition of the integral terms is formed. This augmented state space is as in (9) and (10), respectively.

$$\bar{\mathbf{x}} = \bar{\mathbf{y}} = \begin{bmatrix} \mathbf{e}_x \\ \mathbf{z} \end{bmatrix} \tag{9}$$

$$\begin{aligned} \dot{\bar{\mathbf{x}}} &= \bar{\mathbf{A}}\,\bar{\mathbf{x}} + \bar{\mathbf{B}}\,\mathbf{e}_u, \\ \bar{\mathbf{y}} &= \bar{\mathbf{C}}\,\bar{\mathbf{x}}. \end{aligned} \tag{10}$$

where

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{C} & \mathbf{0} \end{bmatrix}_{4\times 4} \tag{11}$$

$$\bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix}_{4\times 2} \tag{12}$$

$$\bar{\mathbf{C}} = \begin{bmatrix} \mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}_{4\times 4} \tag{13}$$

Once the augmented system states in (9) go to zero, the original system will also track the setpoint x^*. Given the structure of the augmented system, it is straightforward to show that the controllability matrix associated with the augmented system (14) is of full rank, hence the augmented system is controllable.

$$\mathcal{C} = \begin{bmatrix} \bar{\mathbf{B}} & \bar{\mathbf{A}}\bar{\mathbf{B}} & \bar{\mathbf{A}}^2\bar{\mathbf{B}} & \bar{\mathbf{A}}^3\bar{\mathbf{B}} \end{bmatrix} \tag{14}$$

The objective of the controller is to minimize the performance measure defined in (15).

$$J = \frac{1}{2}\int_0^\infty \left(\bar{\mathbf{x}}^\top \bar{\mathbf{Q}}\,\bar{\mathbf{x}} + \mathbf{e}_u^\top \bar{\mathbf{R}}\,\mathbf{e}_u\right) dt \tag{15}$$

In (15), $\bar{\mathbf{R}}$ represents the input penalties, and it is chosen to be a positive definite matrix. $\bar{\mathbf{Q}}$ represents the state penalties and is selected to be a positive semidefinite matrix. According to [21], the optimal input is achieved as in (16),

$$\mathbf{e}_u = -\,\bar{\mathbf{R}}^{-1}\bar{\mathbf{B}}^\top \mathbf{P}\,\bar{\mathbf{x}} = -\,\mathbf{K}\,\bar{\mathbf{x}} \tag{16}$$

where $\bar{\mathbf{P}}$ is obtained by solving the algebraic Riccati equation as shown in (17) and $\mathbf{K}$ is a 2×4 matrix as in (18).

$$\mathbf{P}\,\bar{\mathbf{B}}\,\bar{\mathbf{R}}^{-1}\bar{\mathbf{B}}^\top\mathbf{P} - \mathbf{P}\,\bar{\mathbf{A}} - \bar{\mathbf{A}}^\top\mathbf{P} - \bar{\mathbf{C}}^\top\bar{\mathbf{Q}}\,\bar{\mathbf{C}} = \mathbf{0} \tag{17}$$

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{21} & k_{22} & k_{23} & k_{24} \end{bmatrix} \tag{18}$$

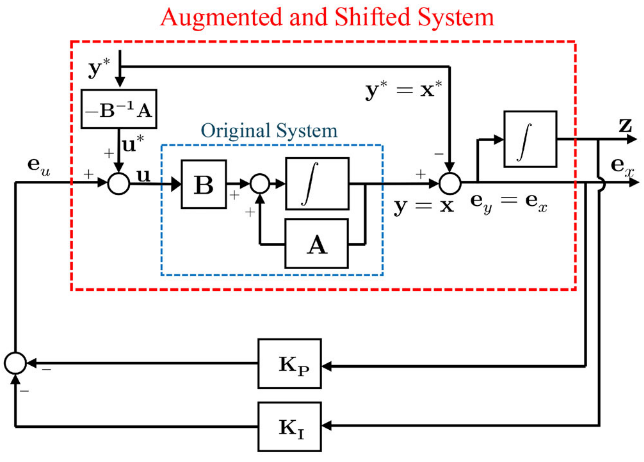

FIGURE 3 | Schematic of the MIMO controller in state space form.

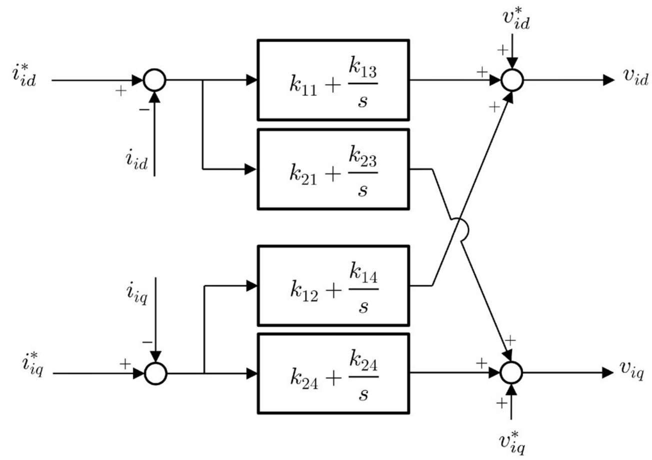

FIGURE 4 | Schematic of the developed current controller for VSCs.

By choosing proper penalties, one can find a desirable response by finding a tradeoff between the control input effort e_u and the speed at which the new states $e_x(t)$ and $z(t)$ go to zero. The advantage of this method is that the penalties, specifically $\bar{\mathbf{Q}}$, can be selected to distinguish between the priority given to the integral gains or proportional gains of the PI controller. In addition, one can prioritize the response of one axis to achieve better transient and steady-state tracking specifications than the other axis. By substituting (9) into (16) and partitioning $\mathbf{K}$ into proportional and integral gain, (19) is obtained.

$$\mathbf{e}_u = -\mathbf{K}_\mathrm{P}\,\bar{\mathbf{x}} - \mathbf{K}_\mathrm{I}\,\mathbf{z} \tag{19}$$

where the proportional and integral controller gains are as in (20).

$$\mathbf{K}_\mathrm{P} = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}, \mathbf{K}_\mathrm{I} = \begin{bmatrix} k_{13} & k_{14} \\ k_{23} & k_{24} \end{bmatrix} \tag{20}$$

By substituting (7) into (19), (21) is obtained, which represents the optimal PI control law from the augmented error-state model.

$$\mathbf{u} = \mathbf{K}_\mathrm{P}\,(\mathbf{x}^* - \mathbf{x}) + \mathbf{K}_\mathrm{I} \int_0^t [\mathbf{x}^* - \mathbf{x}(\tau)]\,d\tau + \mathbf{u}^* \tag{21}$$

Figure 3 shows the schematic of the augmented system and the controller based on the state space form. By augmenting the original current dynamics with the integral of the tracking error, the control design is converted into a regulation problem in which both the instantaneous current error and its accumulated value are driven to zero. Solving the corresponding LQR problem yields a state-feedback law on the augmented state, which can be rewritten in PI form. In this expression, $\mathbf{K}_\mathrm{P}$ and $\mathbf{K}_\mathrm{I}$ are the proportional and integral gain matrices, respectively, while $\mathbf{u}^*$ is the steady-state input required to maintain the desired equilibrium current reference. Since $\mathbf{K}_\mathrm{P}$ and $\mathbf{K}_\mathrm{I}$ are full matrices, each control input can depend on both d- and q-axis errors, allowing the controller to account for the multivariable coupling of the converter dynamics. As shown in Figure 4, its structure closely resembles that of conventional PI controllers shown in Figure 2. As shown in the next section, the MIMO controller significantly improves the dynamic performance and grid synchronization stability of IBRs in various conditions.

TABLE 1 | Test system parameters.

Symbol	Description	Value
R_f	Resistance of the LC filter	20 mΩ
L_f	Inductance of the filter	600 μH
C_f	Capacitance of the filter	12 μF
f_{sw}	Switching frequency of the IBR	5000 Hz
v_{dc}	DC side voltage of the VSC	1000 V
$\mathbf{v}_g$	Nominal voltage of the grid	500 V
S_IBR	Nominal capacity of the IBR	100 kW
k_p	PI current controller's proportional gain	0.13
k_i	PI current controller's integral gain	11.25
k_p^PLL	Proportional gain of the PLL	48
k_i^PLL	Integral gain of the PLL	144
t_d	Control delay	$\frac{3}{2f_{sw}}$
Parameters of the developed MIMO current controller		
$\bar{\mathbf{Q}} = \mathbf{diag}(0.0769,\ 0.0769,\ 70,\ 70)$, $\bar{\mathbf{R}} = \mathbf{diag}(1,1)$		
$\mathbf{K}_\mathrm{P} = \begin{bmatrix} 0.2690 & 0 \\ 0 & 0.2690 \end{bmatrix}$ $\mathbf{K}_\mathrm{I} = \begin{bmatrix} 7.0076 & -4.5710 \\ 4.5710 & 7.0076 \end{bmatrix}$		

4 | Case Study Results

The system, as depicted in Figure 1, is simulated with the parameters listed in Table 1. The delay of $t_d = 1.5/f_{sw}$ is considered in the control loops due to one switching period for the discrete transformation in the digital controller, and half a sample for the PWM operation [24]. The time-domain simulations were conducted in MATLAB/Simulink using an average-value model of the grid-tied VSC, following the modeling framework in [25]. The model includes the filter, a synchronous reference frame (SRF) PLL, and PWM-equivalent control delay, and is used to compare the relative dynamic and synchronization performance

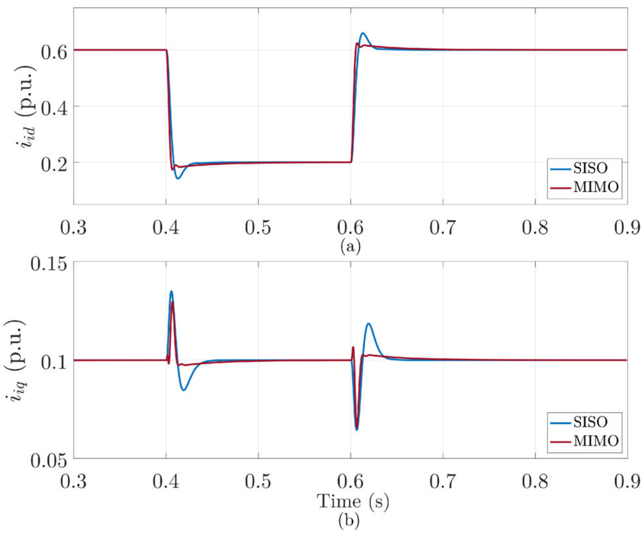

FIGURE 5 | Comparison of controllers' transient responses (a) d-axis (b) q-axis in facing the change in d-axis current reference in a relatively strong grid condition (SCR = 5).

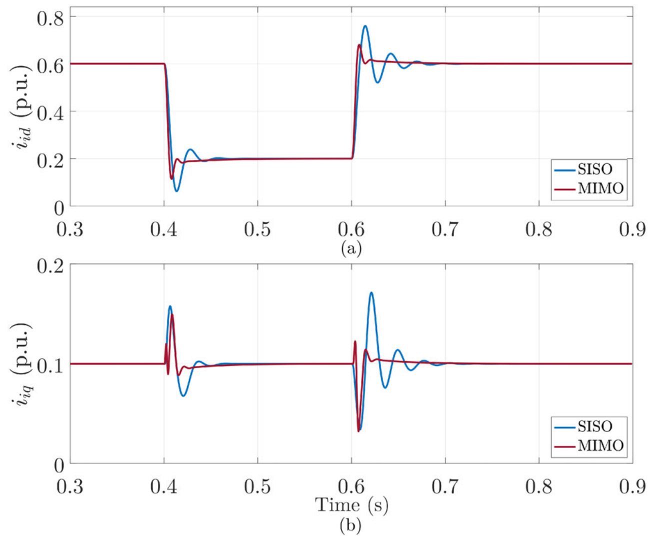

FIGURE 6 | Comparison of controllers' transient responses (a) d-axis (b) q-axis in facing the change in d-axis current reference in an extremely weak grid condition (SCR = 1.95).

of the SISO and proposed MIMO controllers. Several scenarios are studied to demonstrate the advantages of the developed current controller. The matrices $\bar{\mathbf{Q}}$ and $\bar{\mathbf{R}}$ were selected to achieve a tradeoff between fast current tracking, oscillation damping, and control effort. In this work, $\bar{\mathbf{R}} = \mathbf{I}$ was used to penalize both control inputs equally, while $\bar{\mathbf{Q}}$ was chosen as a diagonal matrix with equal weights on the current-error states and higher weights on the integral states. The final values were obtained through iterative tuning of the closed-loop response and were kept unchanged in all relevant case studies. The controller parameters remain unchanged during these scenarios. Also, the benchmark SISO controller was tuned to achieve an appropriate response in terms of percent overshoot, response time, and proper damping, according to [25, 26]. The final gains were then validated in time-domain simulation and kept unchanged throughout the comparative case studies. Additionally, small-signal stability analysis provides an analytical demonstration of the controller's merits. The following subsections discuss the details of the studied cases.

4.1 | Step Changes on the *d*-Axis Set Point

First, the set points of the d and q axes are changed individually to show the system's response to various step-up and step-down changes to the set points.Initially, the grid is relatively strong with short circuit ratio (SCR) of 5 at the point of common coupling of the IBR to the grid, with $X_g/R_g = \tan(80°)$. First, the current setpoint of the d-axis is changed from 0.6 p.u. to 0.2 p.u. and back to 0.6 p.u. at $t = 0.4$ s and $t = 0.6$ s, respectively, while the q-axis setpoint is fixed at 0.1 p.u. As shown in Figure 5, the developed MIMO controller is noticeably superior, with less oscillation and faster tracking response. While it is 2 ms faster, it has 9% less overshoot, and its 5% settling time is 3.3 ms faster. In addition, the developed controller provides a better decoupling of the axes; while the d-axis is changed, the q-axis is affected much less than the conventional SISO PI.

The same scenario of changing the set point is investigated, but SCR is 1.95, representing an extremely weak grid. As shown in Figure 6, the MIMO controller can maintain a desirable response, while the SISO controller oscillates noticeably. PLL dynamics combined with the weak grid's impedance create a strong coupling with the current controller loop [27]. As a result, the ideal proportional decoupling terms in conventional controllers fail to cancel out the coupling entirely. On the other hand, the MIMO controller successfully damps out these oscillations. In summary, the MIMO controller provides superior dynamic performance in both strong and weak grid conditions and better decoupling of the controller's two axes.

4.2 | Step Changes on the *q*-Axis Set Point

The q-axis is directly responsible for reactive power control, which is a crucial factor in maintaining voltage stability in power grids. The injected power to the grid will deviate from the desired setpoints if this response is sluggish, and if oscillatory, the system will be prone to undamped oscillations that may lead to loss of synchronism of the IBR. Additionally, better dynamic response of i_{iq} helps to counteract fluctuations in the voltage, allowing the faster PLL to track the changes in the grid [28]. Figures 7 and 8 show the performance of each controller when a step change is applied on the q-axis in a strong and weak grid, respectively. As shown in these figures, the MIMO controller has a noticeably superior dynamic performance in both cases. The performance improvement of the MIMO controller is more evident in the weak grid scenario.

4.3 | The Effect of Impedance Angles

Equations (22) and (23) represent the static power transfer capability of a grid tied VSC, assuming the inverter has an output voltage $\mathbf{v}_o$ with the angle θ [29].

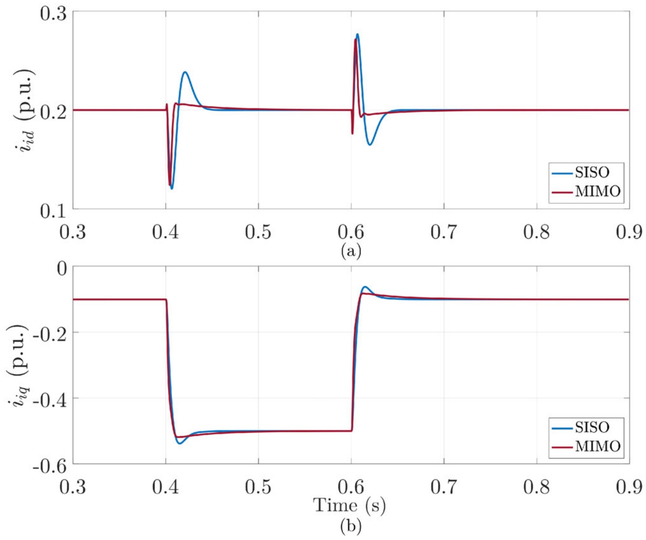

FIGURE 7 | Comparison of controllers' transient responses (a) d-axis (b) q-axis in facing the change in q-axis current reference in a relatively strong grid condition (SCR = 5).

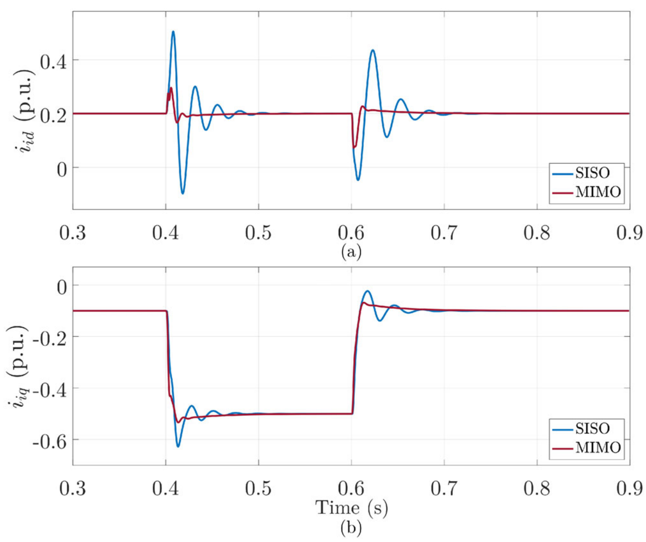

FIGURE 8 | Comparison of controllers' transient responses (a) d-axis (b) q-axis in facing the change in q-axis current reference in an extremely weak grid condition (SCR = 1.95).

$$P_{\text{inv}} = \frac{\mathbf{v}_o \mathbf{v}_g}{|Z_g|} \sin\left(\theta - 90^\circ + \tan^{-1}\frac{X_g}{R_g}\right) + \mathbf{v}_o^2 \frac{R_g}{|Z_g|^2} \tag{22}$$

$$Q_{\text{inv}} = -\frac{\mathbf{v}_o \mathbf{v}_g}{|Z_g|} \cos\left(\theta - 90^\circ + \tan^{-1}\frac{X_g}{R_g}\right) + \mathbf{v}_o^2 \frac{X_g}{|Z_g|^2} \tag{23}$$

Therefore, the static limit of the active power of the grid tied VSC in the p.u. system is shown in (24). In this condition, the injected reactive power is as (25).

$$P_{p.u}^{\max} = \text{SCR}\left(\frac{\mathbf{v}_g}{\mathbf{v}_o} + \frac{R_g}{|Z_g|}\right) \tag{24}$$

$$Q_{p.u} = \text{SCR}\,\frac{X_g}{|Z_g|} \tag{25}$$

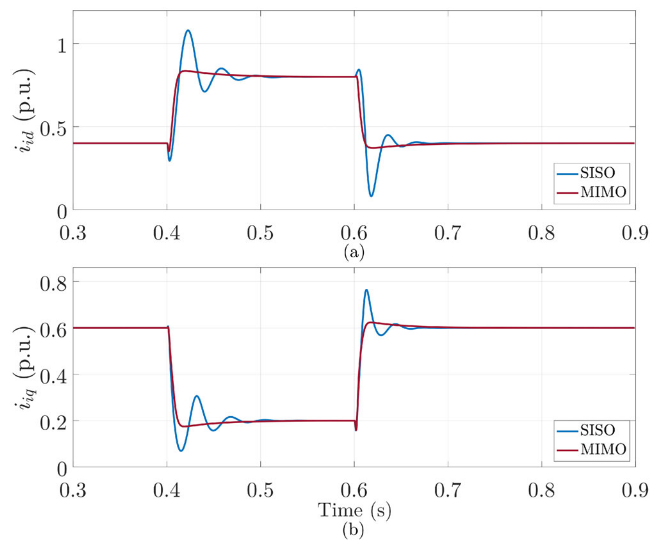

FIGURE 9 | Comparison of controllers' transient responses (a) d-axis (b) q-axis in facing the changes on both d and q axes in an extremely weak grid (SCR = 1) with $X_g/R_g = 1$.

where the SCR is defined as in (26).

$$\text{SCR} = \frac{\mathbf{v}_o^2}{|Z_g|\, S_{\text{IBR}}} \tag{26}$$

In the previous section, $X_g/R_g = \tan(80^\circ)$ was considered. However, some reports have investigated weak grid incidents with a low X_g/R_g ratio as well [30]. Hence, this section investigates the controller's performance in low-impedance angles, where $X_g/R_g = 1$. Figure 9 depicts the response of the controller with this impedance ratio, where simultaneous step changes were applied to the d-axis and q-axis current references at $t = 0.4$ s, and the responses of the SISO and MIMO controllers were compared under the weak-grid condition considered in this subsection.

The dynamic active- and reactive-power limits were obtained from time-domain simulations under the specified grid condition. For each controller, the corresponding current reference was gradually increased, and the system response was monitored. The dynamic limit was identified as the maximum injected active or reactive power for which the converter still maintained a stable response and preserved grid synchronization. Assuming the magnitude of $\mathbf{v}_o$ is close to 1 p.u., according to (24), the static power transfer limit is SCR × 1.707 p.u. However, due to dynamic limits, at SCR = 1, the SISO controller can inject at most 0.94 p.u. of active power before it loses its grid synchronization and becomes unstable in the studied system. The MIMO controller can inject 1.66 p.u. of active power, which means the active power transfer capability has improved by 36%. This is because the MIMO controller provides better dynamic performance, as shown in Figure 9. Also, at SCR = 1, the SISO controller can inject at most 0.49 p.u. of reactive power, but the MIMO controller can go as high as 0.66 p.u. of reactive power, which is 24% higher than that of the conventional SISO controller.

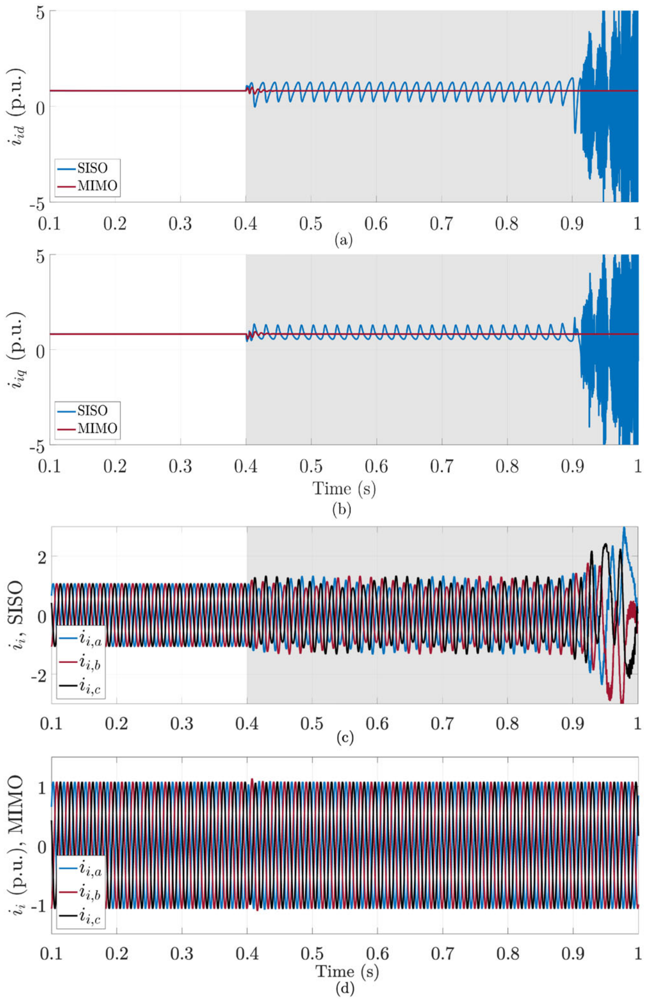

FIGURE 10 | Comparison of the controllers' responses to a sudden change to the grid strength, changing from SCR = 4 to SCR = 2 at $t = 0.4$ s; (a) and (b) represent the currents in SRF, (c) and (d) represent the current of the SISO and MIMO in *abc* frame, respectively.

4.4 | Grid Synchronization Stability Study Using Time-Domain Simulation

This section demonstrates the enhancement of grid synchronization stability by the MIMO controller in response to sudden changes in grid strength. This scenario is motivated by a real-life incident in a solar power plant in Arizona [31]. In that incident, due to outages of a few lines, the strength of the grid dropped, causing oscillatory power injections by the solar power plant. In this subsection, it is assumed that the VSC is interfaced to the grid through two identical transmission lines. Initially, the grid strength is SCR = 4, with an impedance angle of 80 degrees. The VSC injects approximately equal active and reactive powers of 0.66 p.u. This is well below the static limit of $P_{p.u}^{max} = 4.69$ p.u. and its corresponding reactive power of $Q_{p.u} = 1.67$ p.u., calculated based on (24) and (25). At $t = 0.4$ s, one of the lines is taken out, dropping the SCR to 2. Following this event, as shown in Figure 10c, the SISO controller fails to synchronize to the system. On the other hand, as shown in Figure 10d, the MIMO controller maintains synchronism, which shows its superior ability to maintain the grid synchronization stability. In addition, Figure 11 shows the PLL angle dynamics during this scenario,

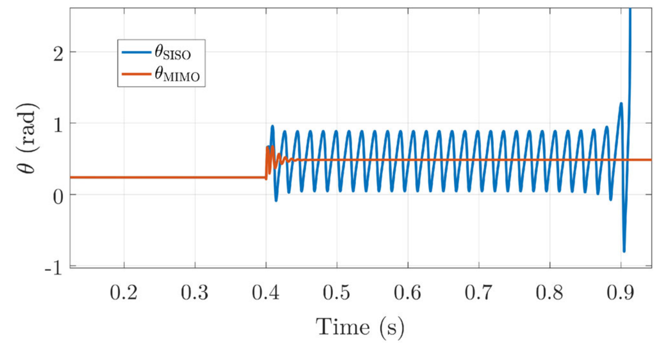

FIGURE 11 | PLL angle dynamics following the line outage scenario in Section 4.4.

where the SISO controller fails to maintain synchronism, and the output angle keeps increasing.

4.5 | Grid Synchronization Stability Study Using Analytical Analysis

In this section, eigenvalue analysis is performed to study the grid synchronization stability enhancement by the MIMO PI controller. The analytical study in this subsection is based on the state-space model of the closed-loop grid-tied converter. The nonlinear model is linearized around the operating point, and the state matrix $\mathbf{A}_{sys}$ is obtained as the Jacobian of the state equations with respect to the state vector. The eigenvalues of $\mathbf{A}_{sys}$ are then used to assess grid-synchronization stability. The corresponding state-space model for the SISO case is summarized in [32]. The state-space structure is the same for both controllers, and the difference lies in the current control laws. Specifically, the conventional SISO controller uses (27), whereas the MIMO controller uses (28).

$$\begin{cases} v_{id} = k_p\left(i_{id}^* - i_{id}\right) + k_i\gamma_d - \omega_s L_f i_{iq} + v_{od}, \\ v_{iq} = k_p\left(i_{iq}^* - i_{iq}\right) + k_i\gamma_q + \omega_s L_f i_{id} + v_{oq}. \end{cases} \tag{27}$$

$$\begin{cases} v_{id} = k_{11}\left(i_{id}^* - i_{id}\right) + k_{12}\left(i_{iq}^* - i_{iq}\right) + k_{13}\gamma_d + k_{14}\gamma_q + v_{id}^* + v_{od}, \\ v_{iq} = k_{21}\left(i_{id}^* - i_{id}\right) + k_{22}\left(i_{iq}^* - i_{iq}\right) + k_{23}\gamma_d + k_{24}\gamma_q + v_{iq}^* + v_{oq}. \end{cases} \tag{28}$$

Figure 12 shows the system's eigenvalues for systems with different grid strength, where Figure 12a shows the results for the conventional SISO controller and Figure 12b shows the results for the MIMO controller. The system eigenvalues are plotted by sweeping through strength values from SCR = 4 to SCR = 2, observing their corresponding changes. It is evident that for the SISO controller, two of the eigenvalues soon move towards the right half-plane, causing the system to become unstable. In contrast, the MIMO controller remains stable even under weaker conditions.

The grid synchronization stability of the VSC under parameter variations is also investigated. Figure 12 shows the eigenvalues for 50 different values of the resistor and inductor of the filter. These values are selected from a range that spans from the nominal

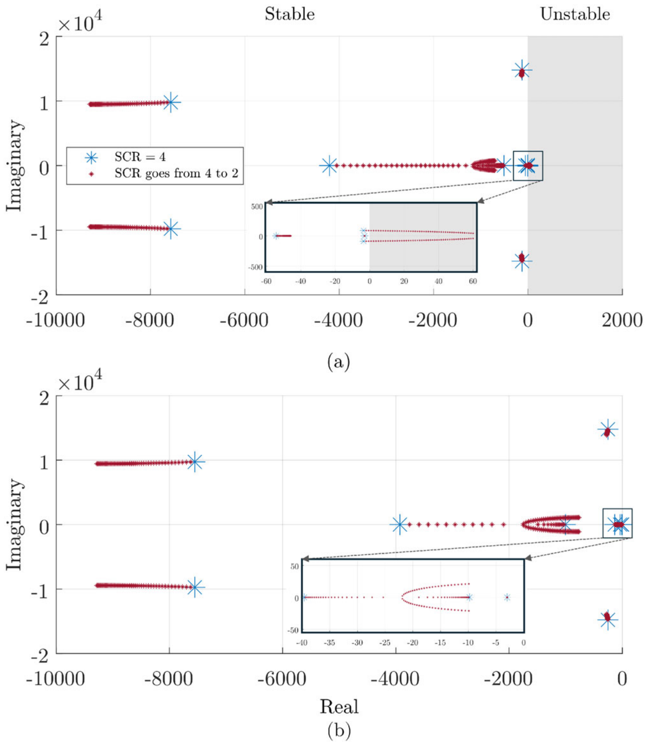

FIGURE 12 | Comparison of eigenvalues of the system with (a) SISO-PI (b) MIMO-PI controllers at different grid strength values.

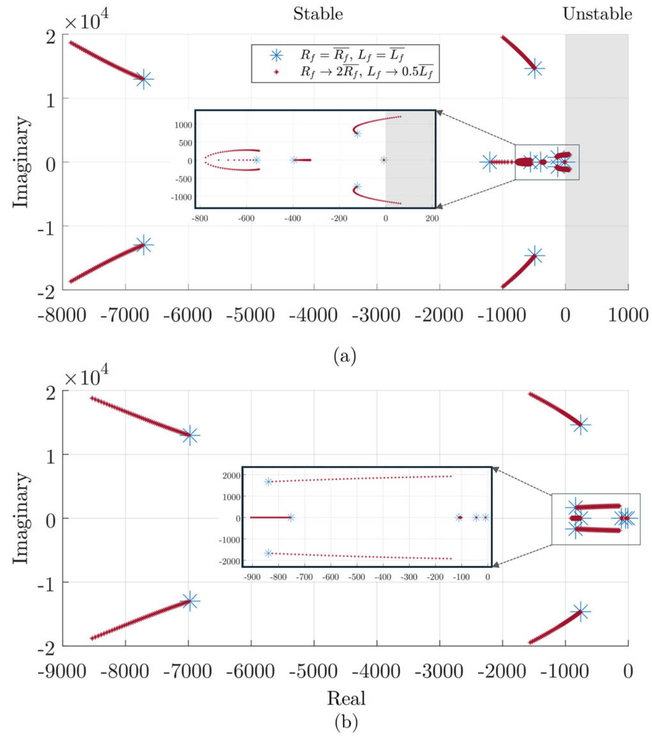

FIGURE 13 | Comparison of the eigenvalues in (a) SISO-PI and (b) MIMO-PI controllers by changing the filter parameters of the VSC.

value used in the controller design to twice the nominal value for the resistor and half the nominal value for the inductor. The grid strength is set at SCR = 2. As shown in Figure 13, the eigenvalues in the MIMO-PI controller move toward the right half-plane at a slower rate compared to those in the SISO controller.

4.6 | Multi-Machine Response Study

In this section, the performance of the MIMO controller is evaluated using the power system shown in Figure 14. The system consists of two synchronous generators, each equipped with an AVR and a turbine governor. The turbine governor was modeled using the IEEEG1 governor structure, with some parameters neglected for simplification, following the implementation in [32] and the standard IEEEG1 model in [33]. The excitation system was represented by a simplified lead-lag plus first-order exciter model following the implementation in [32]. For the IBR system, two transmission lines with relatively high impedances are used to represent a weak interconnection. The IBR parameters are also selected for this system's rating, and all the parameters are listed in the Appendix.

First, the system performance is examined under a relatively weak-grid condition. The VSC is initially set to inject 2.5 MW and 2.83 Mvar of active and reactive power, respectively. In the first scenario, a 0.2 p.u. step disturbance is applied to i_d^* at $t = 10$ s. As shown in Figure 15, the MIMO controller provides significantly improved rise time and settling time compared with the SISO controller, while having a similar overshoot.

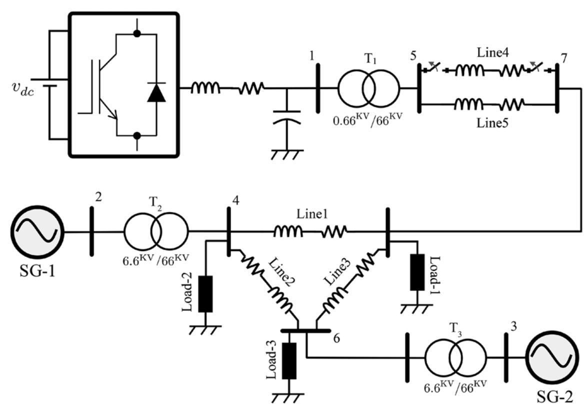

FIGURE 14 | Multi-machine power system under study [34, 35].

In the second scenario, Line-4 is disconnected at $t = 10$ s. The synchronous generators' rotor frequency and the IBR's estimated PLL frequency dynamics are shown in Figure 16. As demonstrated in Figure 16, the MIMO controller is able to damp the resulting weak-grid oscillations, whereas the SISO controller loses synchronism.

Finally, in Scenario 3, a frequency disturbance is introduced by decreasing SG-1's governor reference by 35% at $t = 10$ s. Since the loads remain unchanged, the active power shortage creates a decrease in system frequency. For the VSC, the responses of the two controllers are examined while the current references are kept unchanged. As shown in Figure 17, both controllers maintain satisfactory performance.

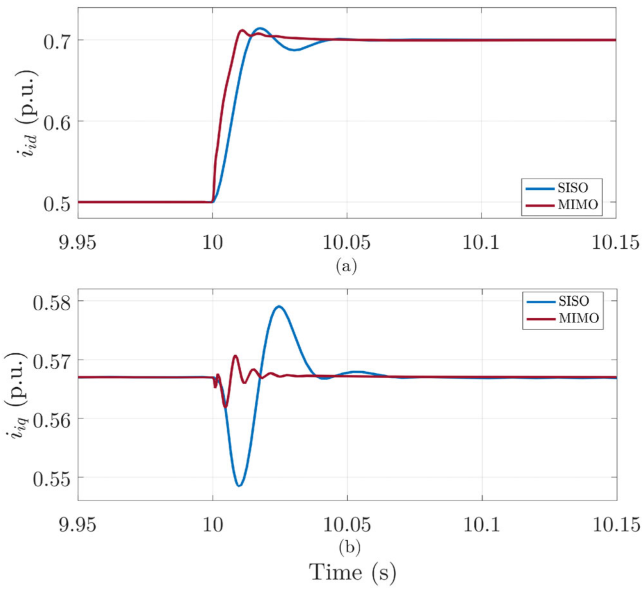

FIGURE 15 | Comparison of the controllers' transient responses (a) d-axis (b) q-axis in facing the change in d-axis current reference in a relatively weak interconnection.

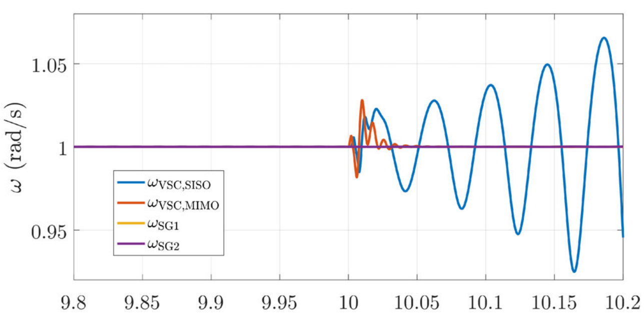

FIGURE 16 | Rotor/PLL frequency dynamics of the sources in response to Line4 outage at $t = 10$ s.

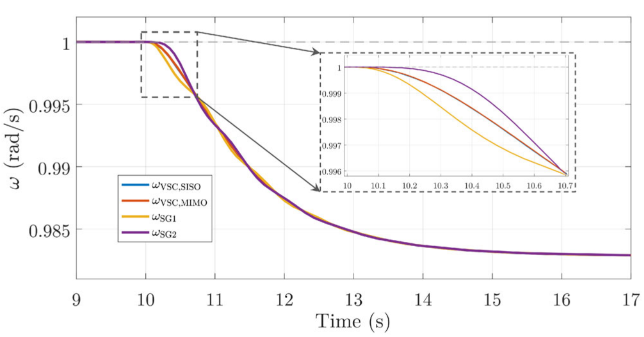

FIGURE 17 | Rotor/PLL frequency dynamics of the sources in response to SG-1's governor reference change by 35% at $t = 10$ s.

5 | Conclusion

This paper developed a current controller strategy for VSCs based on optimal control theory. The proposed method leverages the simplicity of the SISO PI controller and preserves its familiar structure, which is well known to VSC controller designers and power system operators engaged in IBR grid-integration studies. It does not require a complete redesign of conventional controllers to replace them with nonlinear controllers. The method provides improved decoupling performance and robustness against system-parameter variations, such as line outages.

In designing the controller, practical limitations are explicitly considered. For instance, while some approaches improve controller response by integrating PLL dynamics into the control design process, the developed controller does not explicitly include PLL dynamics. As a result, it remains compatible with various PLL types and structures, and the current controller does not need to be redesigned for different PLL implementations.

Overall, the proposed controller enhances system response under both strong- and weak-grid conditions, as well as during contingencies, without requiring additional control layers or corrective measures. Thus, the method improves robustness while maintaining the practical simplicity of conventional PI control. Time domain simulation results and analytical analysis demonstrated enhanced grid synchronization stability, decoupling performance, and power transfer capability of the developed controller.

Nomenclature

Parameters

$\mathbf{v}_i = [v_{id}, v_{iq}]$	dq-axis input voltage of the VSC
$\mathbf{v}_o = [v_{od}, v_{oq}]$	dq-axis output voltage of the VSC
$\mathbf{v}_g = [v_{gd}, v_{gq}]$	dq-axis Thevenin voltage of the grid
R_f, L_f, C_f	Filter parameters of the VSC
R_g, L_g, Z_g	Thevenin impedance of the grid
$\mathbf{i}_i = [i_{id}, i_{iq}]$	dq-axis input current of the VSC
$\mathbf{i}_o = [i_{od}, i_{oq}]$	dq-axis current injected into the grid
$\mathbf{A}, \mathbf{B}, \mathbf{C}$	Original system state-space matrices
$\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{C}}$	Augmented system state-space matrices
$\mathbf{x}^* = [i_{id}^*, i_{iq}^*]$	Set-points of the dq-axis input current
$\mathbf{u}^* = [v_{id}^*, v_{iq}^*]$	Constant inputs corresponding to x^*

Controller parameters

$\mathbf{z}(t)$	Integral state of the PI controller
k_p, k_i	PI gains of the SISO current controller
$k_p^{\mathrm{PLL}}, k_i^{\mathrm{PLL}}$	PLL controller gains
$\mathbf{K}, \mathbf{K}_P, \mathbf{K}_I$	Optimal PI controller gains
$\bar{\mathbf{Q}}, \bar{\mathbf{R}}$	State and input penalty matrices of the LQR
$J, \bar{\mathbf{P}}$	LQR cost function and Riccati solution

Author Contributions

Hassan Yazdani: conceptualization, data curation, methodology, validation, visualization, writing – original draft, review, and editing. **Ali Maleki:** conceptualization, data curation, investigation, methodology, validation, writing – review and editing. **Saeed Lotfifard:** conceptualization, data curation, investigation, methodology, project administration, resources, supervision, validation, visualization, writing – original draft, review, and editing. **Ali Saberi:** conceptualization, investigation, methodology, supervision, validation, writing – original draft, review, and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Parameters of the Multi-Machine System

The transformers are modeled with a Yg/Yg connection and a rated power of 6 MVA. The synchronous machine parameters are:

$S_{rate} = 5.0\,\text{MVA}$, $V_{rate} = 6.6\,\text{kV}$, $H = 8$, $D = 0\,\text{p.u.}$, $R_s = 0.0025\,\text{p.u.}$, $X_d = 0.296\,\text{p.u.}$, $X_q = 1.06\,\text{p.u.}$, $X_d' = 0.296\,\text{p.u.}$, $X_q' = 0.296\,\text{p.u.}$, $X_d'' = 0.177\,\text{p.u.}$, $X_q'' = 0.177\,\text{p.u.}$, $X_l = 0.2\,\text{p.u.}$, $T_{d0}' = 3.7\,\text{s}$, $T_{q0}' = 0.5\,\text{s}$, $T_{d0}'' = 0.05\,\text{s}$, $T_{q0}'' = 0.05\,\text{s}$, $K_g = 10.0\,\text{p.u.}$, $P_0 = 5\,\text{MW}$, $T_1 = 0.05\,\text{s}$, $T_2 = 0\,\text{s}$, $T_3 = 0.15\,\text{s}$, $T_4 = 0.08\,\text{s}$, $T_5 = 0.15\,\text{s}$, $K_1 = 0.6\,\text{p.u.}$, $K_2 = 0.4\,\text{p.u.}$, $T_A = 0.1\,\text{s}$, $T_B = 0.01\,\text{s}$, $K_e = 50.0\,\text{p.u.}$, $T_e = 1\,\text{s}$.

The VSC parameters are:

$S_{rate} = 5.0\,\text{MVA}$, $V_{rate} = 0.66\,\text{kV}$, $R_f = 0.0014\,\text{ohm}$, $L_f = 0.1387\,\mu\text{H}$, $C_f = 25.4\,\text{mF}$, $\omega_s = 2\pi 60\,\text{rad/sec}.P_0 = 2.5\,\text{MW}$, $Q_0 = 2.835\,\text{MVar}$, $K_p = 0.0028 K_i = 0.2773 K_p^{PLL} = 20\,\text{p.u.}$, $K_i^{PLL} = 100\,\text{p.u.}$, $\bar{\mathbf{Q}} = \mathbf{diag}(0.05,\ 0.05,\ 20,\ 20)$, $\bar{\mathbf{R}} = \mathbf{diag}(1,\ 1)$

$$\mathbf{K}_\text{P} = \begin{bmatrix} 0.0702 & 0 \\ 0 & 0.0702 \end{bmatrix}, \quad \mathbf{K}_\text{I} = \begin{bmatrix} 4.4603 & -0.3257 \\ 0.3257 & 4.4603 \end{bmatrix}.$$

The load/transmission line parameters are:

Load-1= $2.5\text{MW} + j1\text{MVar}$, Load-2= $5\text{MW} + j1.5\text{MVar}$, Load-3= $5\text{MW} + j1.5\text{MVar}$,

$Z_\text{Line1} = 0.15 + j1.47\Omega$, $Z_\text{Line2} = 0.1 + j0.98\Omega$, $Z_\text{Line3} = 0.1 + j0.98\Omega$, $Z_\text{Line4} = Z_\text{Line5} = 20 + j192\Omega$, $Z_\text{T1} = Z_\text{T2} = Z_\text{T3} = 0.1\text{p.u.}$,